

Nonprofit Growth-Signal Engine

An enrichment waterfall for identifying live buying signals — and only paying for enrichment on the shortlist that earns it.

Take-Home Exercise · GTM Engineering · GTJM Systems

Part One — Executive Brief

The Ask

A sales leader wants every mid-to-large nonprofit actively hiring a **Director of Development** or **Development Ops** role right now — found, scored by urgency, and routed to the right AE fast.

The Approach

The system runs in three phases, and each phase spends money differently:

- **Collect.** A cron-scheduled scraper and a Cowork LinkedIn-alert parser feed one deduped JSON ledger. This layer is nearly free — no paid enrichment runs here.
- **Score.** Every posting collapses to one row per company; each company is scored against peers of similar size, and only statistical outliers — hiring beyond normal churn — clear the bar.
- **Expand.** Only the flagged shortlist fans out to contacts — deduped, enriched, and routed to CRM or Gong. This is the only layer where paid credits are spent.

The Shape of the System

Data volume in this pipeline looks like a barbell: many raw postings narrow down to one row per company, get scored down to a small outlier shortlist, and then fan back out to multiple contacts per flagged company. Two points in that shape are where duplicates are made or prevented:

- **The narrow neck (Postings → Companies):** every posting has to prove itself against the CRM — by name, then by enriched domain and LinkedIn URL — before it is allowed to create or attach to a Company record. Skip this and one org with three concurrent postings becomes three duplicate companies.
- **The wide end (Companies → Contacts):** one flagged company can surface several target-title contacts, and the same person can surface twice if more than one posting triggered the search. Contacts are deduped on email and LinkedIn URL, and matched against the CRM, before any paid enrichment runs.

Key Decisions

- **Match to CRM before spending credits, at both the Account and Contact level.** This is the single rule that prevents duplicate Accounts and duplicate Contacts from entering Salesforce.

- **Score against peer tiers, not the raw population.** A flat count of open postings just re-ranks the biggest organizations. Scoring within employee-count bands (Small / Mid / Large) isolates real hiring intensity from ordinary staffing churn.
- **Reserve paid discovery for the shortlist, not the whole segment.** A scraper plus a cron job is nearly free to run and only needs occasional maintenance when a site's HTML changes. Paid enrichment (Clay credits) is spent only on companies that already cleared the outlier bar.
- **The signal is live, not just leading.** A specific job posting can close or expire, but the target-title contacts at a flagged organization remain real, callable people at a company that has demonstrably invested in fundraising capacity — the lead does not go stale when one req fills.

Bottom line

This system is designed to answer one question honestly: which nonprofits are investing in fundraising capacity right now, beyond what normal turnover would explain — and who, specifically, should an AE call about it. Everything after the Collect phase exists to protect that answer from noise, duplication, and unnecessary spend.

Part Two — Technical Deep Dive

1. Data Collection

Sources: **Idealist**, **AFP Career Center**, state nonprofit association boards (California, Minnesota, Maryland, and similar), and **LinkedIn job-alert emails** from saved searches.

- **Claude Code scraper** runs on a cron job against the nonprofit-specific job boards.
- **Cowork task** parses LinkedIn's saved-search alert emails via the connected Gmail account.
- **Dedupe logic** matches on posting URL, or on organization + title + post date when no stable URL exists.
- **JSON Database** is the single source of truth. New records are deduped here before anything is sent onward — this keeps day-to-day scraping runs free, since storage and comparison cost nothing per record.

2. The Postings Table (Clay) — First Weight of the Barbell

New JSON records are POSTed to a Clay webhook, creating one row per posting in the Postings Table. Clay webhook sources cap at 50,000 submissions per table — a non-issue at this volume, but worth flagging if the source list ever expands beyond nonprofit boards.

Enrichment waterfall (runs inside the Postings Table):

Step	Action	Trigger	Outcome
1	Match to CRM by Company Name	Runs on every new row	Match found → attach signal to existing Account (done). No match → continue to Step 2.
2	Clay Native Enrich (1 credit)	Only if Step 1 found no match	Returns Domain, normalized Company Name, LinkedIn URL, Employee Count.
3	Retry CRM Match	Runs after Step 2	Cascades Domain → LinkedIn URL → Name. Match found → attach to existing Account. No match → flag row for new Account creation.

Why the retry step exists: a name-only match misses on abbreviations, punctuation, and “Inc.”-style variants. Clay Native enrichment fills in a domain and a LinkedIn URL — both far more reliable match keys — so the second CRM match attempt isn't working from a fuzzy string alone.

3. The Companies Table — The Barbell's Handle

The Postings Table is deduped on normalized domain (keep the newest row), producing the Companies Table: one row per organization, regardless of how many concurrent postings it has.

Rolling up posting counts: the Companies Table runs a **Lookup Multiple Rows** action — a self-join back to the Postings Table, matched on domain. This returns, for each company, every Postings row sharing that

domain. A formula on the length of that returned list is the count. Two counts are tracked separately: open Director of Development postings, and open Director of Development Operations postings.

Mechanics note: Clay's built-in Dedupe action is exact-match only (case- and whitespace-sensitive), so it will not catch “Acme Inc” vs. “Acme, Inc.” on its own. Normalizing the company name and domain into a shared key has to happen *before* the Lookup Multiple Rows call — the normalization is the load-bearing step, not the lookup itself.

Splitting by size: once counts are attached, the Companies Table is split into three separate tables by employee count — Small (under 50 employees), Mid (50–500), and Large (500+). Each becomes the input to its own Scoring Table in the next stage. (Only Medium to Large for this exercise)

4. The Scoring Table — Finding the Signal

Each employee-count table (Mid, Large) is scored independently, so an organization is only ever compared to its real peers — never against the whole segment at once.

Method	Role	Logic
Median + IQR within tier	Primary	Flags organizations whose concurrent open postings (trailing 90 days) exceed $Q3 + 1.5 \times IQR$ for their tier. Robust to the right-skew that small nonprofit-hiring samples typically have.
Percentile rank within tier	Fallback	Used when a tier's sample is too thin for IQR to be meaningful. Organizations are ranked by percentile of postings within that tier instead.

Converting to a 1–5 score: under the IQR method, any organization flagged above the outlier band automatically scores a 5. Under the percentile method, scores are assigned in bands of 20% (1st–20th percentile = score 1, ..., 81st–100th percentile = score 5).

Trigger condition

If an organization scores a 5: it triggers Find Contacts at Company, searching for target titles. Results are written to the Contacts Table.

5. The Contacts Table — The Barbell's Other Weight

5a. Intake, Dedupe, and CRM Match

Find Contacts at Company searches for target titles — buyer-persona roles at the flagged organization, such as VP of Revenue Operations, Director of Development, or Head of Development Operations — not necessarily the exact requisition that triggered the flag. Results land in the Contacts Table as raw rows.

Contact-level dedupe runs immediately, keyed on email plus LinkedIn URL. This matters because one flagged company can surface several target-title contacts, and the same person can appear twice if more than one posting at the org triggered the search. The same principle from the Postings Table applies here: match first, spend credits second.

After dedupe, each contact is checked against the CRM to determine which enrichment path it takes next.

5b. Enrichment Waterfall and Routing

Branch	Steps	Rationale
1. Match found in CRM	Clay Native LinkedIn → Email waterfall → Direct Phone waterfall	An already-recognized contact justifies the full paid cascade: profile confirmation, then a verified email, then a dial-ready number.
2. No match in CRM	Clay Native LinkedIn → Email waterfall → Direct Phone waterfall + Basic Contact Enrich (location, title, seniority)	A net-new, contact gets enriched with additional information for routing. Then they are
3. Retry CRM Match	Retry CRM match with more data	Cascades email → LinkedIn URL → Name. Match found → attach to existing contact. No match → flag row for new contact/lead creation.

Both branches converge into a single outreach-ready record, routed to a CRM Campaign or a Gong Engage flow.

6. Design Rationale — Objections Addressed

Why not just score by number of openings?

A raw count just re-ranks the biggest organizations. It rewards size, not intent — exactly the vanity metric this system is built to avoid.

Why not have Clay find the postings directly?

Discovery at scale burns paid credits on records that mostly won't clear the bar. A scraper is nearly free to run; it only needs an occasional fix in Claude Code when a source site changes its HTML structure.

Why are there so many CRM checks when we have dedupe rules?

Clay runs through API to CRM and will typically ignore dedupe rules on object creation. CRM checks lower enrichment costs. The goal of this workflow is finding an opportunity not mass enrichment.

7. A Note on Timing — Why This Is a Live Signal

This is a live signal, not just a leading one. A specific requisition can fill or expire, but the target-title contacts surfaced by this system are real people at a real, currently-investing organization — callable today regardless of that one posting's status.

- It may not be the Director of Development or Dev Ops hire who ultimately picks up the phone — it may be another target title who ultimately gets passed through to the CRM for outreach.
- What's real and current is the organization's demonstrated investment in fundraising capacity, not the literal job requisition.
- A filled posting does not invalidate the lead: the Account already cleared the outlier bar on the Scoring Table, and its contacts remain worth calling.

Where freshness still matters: the messaging, not the lead. It's worth re-checking whether a specific posting is still live before referencing it directly in outreach — the lead stays valid either way, but the talk track should match reality.